

Learning with machines: Toward a theory of epistemic co-agency

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ABSTRACT

The rapid integration of generative AI (GenAI) systems into educational settings necessitates a reevaluation of how learners reason, construct, and assume responsibility for knowledge. Existing learning theories like constructivism, sociocultural theory, and connectivism presume human-centered epistemic agency and fail to account for the ways GenAI simulates reasoning, reframes arguments, and co-constructs meaning. This paper introduces the Epistemic Entanglement Framework, a theory-informed model that captures how learners engage with AI systems. Drawing on distributed cognition, sociomaterialism, and posthumanist theory, the framework proposes epistemic co-agency as a reflexive stance wherein learners engage AI outputs dialectically—challenging assumptions, surfacing contradictions, and asserting epistemic sovereignty. Instructional and assessment implications are discussed, emphasizing the need to design for epistemic transformation, not just task efficiency. Ultimately, this work argues that the central challenge of AI in education is not technological fluency, but the cultivation of learners who can reason with, through, and against generative systems. As AI becomes a durable presence in learning, education must focus not only on what students know, but on how they engage with GenAI systems to create knowledge.

Generative artificial intelligence (GenAI) systems have evolved beyond rudimentary versions of grammar checkers (Grammarly) and automated graders (ETS e-rater) (Chen et al., 2016; O'Neill & Russell, 2019). Large language models (LLMs), such as ChatGPT, Claude, and Gemini, are increasingly integrated into learners' daily workflows, offering feedback on writing, generating explanations, and even proposing solutions in problem-based learning environments (Bernabel et al., 2023). These GenAI systems are not static repositories of information like webpages or textbooks. Instead, they generate unique responses, simulate argumentative moves, and influence learners' reasoning trajectories (Wu et al., 2025), which were previously considered primarily human functions. This shift demands a rethinking of how learners reason, assess, and claim knowledge in environments where technologies now perform many of the same cognitive tasks once reserved for humans.

GenAI systems raise epistemological questions about how learners construct, evaluate, and take ownership of knowledge in environments where nonhuman systems now participate in meaning-making. Current learning theories, while foundational, offer limited guidance for this terrain. Constructivist approaches to learning emphasize internal meaning-making and assume human-centered epistemic agency, or one's ability to ask questions, interpret data, and generate knowledge

(Sjøberg, 2010). Sociocultural theories, such as Vygotsky's model of mediated cognition, incorporate tools and artifacts into the learning process; however, there is an intrinsic asymmetry between humans and tools (Kozulin, 2002). Connectivism addresses the distributed and networked nature of digital knowledge while maintaining the primacy of humans in epistemic agency (Siemens, 2017). While these theories remain analytically useful, they are insufficient to guide the design of learning systems in which AI acts not only as a scaffold but also as an epistemic presence.

What these theories share is a foundational assumption: that agency, understanding, and meaning-making are exclusively human capabilities (Hutchins, 2000). They are built on the divide between minds and tools, subjects and objects. However, GenAI systems blur that divide, as they now participate in epistemic processes without agency or intent. When learners engage with systems that generate persuasive, responsive, and strategic knowledge claims, the boundary between tool and co-thinker becomes porous. We cannot meaningfully design for these environments without rethinking the epistemic architecture of learning itself. In this paper, I propose a theory-informed framework for understanding human-AI collaborations in AI-mediated learning.

My central premise is that human-AI collaborations should not be treated as a user-interface problem or as a matter of optimizing tool

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interaction. I contend that human-AI collaborations should be framed as a site of epistemic development where humans cultivate the ability to reason, reflect, and respond critically within hybrid human-machine environments. Philosophical analyses of AI-mediated authorship similarly characterize generative interaction as a distributed epistemic process in which human agents retain evaluative responsibility while delegating generative labor to computational systems. Such activity represents a reconfiguration rather than displacement of epistemic agency, situating knowledge production within hybrid human-machine ecologies (Rodrigues, 2025). This perspective reinforces the importance of conceptual lenses capable of distinguishing forms of epistemic participation rather than treating AI-supported engagement as uniform.

Therefore, I propose the Epistemic Entanglement Framework, structured into three epistemic configurations: augmentation, integration, and co-agency, that describe how reasoning is enacted across human-AI systems. Each configuration reflects a distinct human posture, cognitive demand, and design implication.

For clarity, I use the term epistemic entanglement to refer to the condition in which knowledge construction emerges through reciprocal interaction between human reasoning and AI-generated contributions such that the boundaries of authorship, authority, and justification become difficult to disentangle. In entangled settings, learners do not simply use tools to access information; their judgments, interpretations, and epistemic positioning are shaped through ongoing negotiation with system outputs that influence what appears plausible, persuasive, or credible. This concept provides the motivating premise for the framework developed in this paper. Drawing on theories from sociocultural learning and situativity, the epistemic entanglement framework offers a new lens for designing learning environments that support not only performance but epistemic maturity.

1. Current learning theories and AI-mediated learning

While constructivist, sociocultural, and connectivist theories have shaped educational design and analysis, they fall short in accounting for the epistemic dynamics introduced by generative AI. These theories presuppose that learners are the sole or primary agents of knowledge creation in the learning process (Hollan et al., 2000). When systems like ChatGPT or Claude begin generating novel arguments, reframing problems, or shaping learners' responses in real time (Grande et al., 2024), these foundational assumptions become increasingly unstable.

Constructivism assumes that knowledge is actively constructed by individuals through internal cognitive processes. It emphasizes learner autonomy, sense-making, and reflection as routes to understanding (Sjöberg, 2010). But constructivism provides no account of what happens when a learner outsources part of this construction to a system whose reasoning process is neither transparent nor human. For example, when a student prompts a GenAI to explain a concept and then internalizes that explanation without understanding how it was generated, we are no longer witnessing traditional constructivist learning. The learner's "understanding" is scaffolded by an opaque, probabilistic model. Yet the theory gives us no vocabulary to interrogate this shift. Constructivism offers no guidance for how learners should evaluate or position themselves in relation to such epistemic assistance.

Sociocultural theories, particularly Vygotskian models of mediated cognition, offer greater flexibility by incorporating tools, symbols, and social interaction into their conception of learning (Kozulin, 2002). However, they rely on a clear asymmetry between human agents and mediating artifacts. Tools mediate; humans intend. But GenAI systems blur this boundary. They generate strategic responses, simulate dialogue, and adapt to context without conscious or intentional effort (Wu et al., 2025). For instance, when a student revises their thesis in response to an unexpected counterpoint proposed by ChatGPT, the AI has altered the learner's reasoning path. Yet sociocultural theory lacks a robust account of how unintentional, yet responsive, systems contribute to the co-construction of knowledge.

Connectivism, often cited as a learning theory for the digital age, explicitly acknowledges the distributed and networked nature of contemporary knowledge systems. It allows for non-human nodes in a learning network, such as webpages, databases, and algorithms. However, it still places the learner in the role of epistemic filter: selecting, curating, and connecting information. It assumes that while knowledge may be distributed, reasoning remains a human capacity (Siemens, 2017). But GenAI complicates these assumptions. When students rely on AI to draft, compare, or reframe their arguments (Zhang et al., 2023), they are not merely selecting information. They are engaging in epistemic interaction with a system whose influence is not fully under their control. Connectivism describes distribution, but it cannot fully articulate how learners should navigate the recursive entanglement of reasoning processes that emerge in real-time dialogue with AI systems.

Advocates of these theories may argue that they are flexible enough to incorporate AI. Constructivists might claim that AI outputs are simply another learning resource. Sociocultural theorists might suggest that GenAI is a tool akin to calculators or textbooks. Connectivists might point to the longstanding presence of non-human actors in learning networks. But these rebuttals overlook a critical distinction: GenAI does not merely deliver information; it participates in the process of meaning-making. It generates responses that shape what learners consider plausible, persuasive, or authoritative—often without learners fully grasping why (Clark et al., 2025). These interactions alter not just what is known, but how knowledge is constructed and by whom.

While constructivist, sociocultural, and distributed cognition perspectives offer important foundations for understanding mediated and distributed learning, they do not explicitly center the epistemic positioning required when learners engage with generative systems that produce persuasive knowledge claims. As learners increasingly co-construct ideas with systems that generate, reframe, and provoke without being accountable, we need new frameworks for understanding epistemic activity. Rather than replace existing traditions, the framework proposed here extends them by foregrounding how learners evaluate, justify, and assume accountability for knowledge claims emerging within human-AI interaction.

2. Epistemic agency in AI-mediated learning

Epistemic agency refers to the learner's ability to ask questions, generate explanations, make sense of information, and decide what to believe or challenge (Nieminen & Ketonen, 2024). Beyond reasoning, epistemic agency is the power to shape the knowledge production and practices of a community (Stroupe, 2014; Wu et al., 2025). Miller et al. (2018) emphasize the learners' ability to negotiate meaning and epistemic authority within social and disciplinary contexts.

GenAI complicates this construct. Learners are no longer interacting solely with human teachers, texts, or peers, but with systems that simulate reasoning, produce persuasive explanations, and offer responses that appear authoritative. These systems do not merely transmit information; they actively generate it, often with fluency and coherence that exceed human expectations. As Oh and Lee (2024) argue, models like ChatGPT are epistemic agents "capable of generating new scientific knowledge" (p. 779) whose outputs are treated by learners as plausible, even credible, knowledge claims. Several researchers, however, have begun to diagnose the risks associated with the notion of GenAI as epistemic agents. Rosenberg (2023) warns that conversational AI systems function as personalized, interactive, and strategic influence engines, capable of shaping user beliefs through feedback loops that blur the line between assistance and manipulation. Hauswald (2025) describes large language models as "artificial epistemic authorities" (p. 1), arguing that they exert epistemic influence despite lacking beliefs, intentions, or accountability. From a design standpoint, Alvarado (2023) proposes that GenAI systems be understood as epistemic technologies whose affordances directly shape epistemic practices such as inference, synthesis, and justification. In each case, the concern is that learners

may engage with AI not as a “collaborator in knowledge construction,” but uncritically, defaulting to passive acceptance rather than epistemic interrogation.

If learners are to develop as autonomous and critical knowers, they must not only use AI systems but also learn how to position themselves in relation to them. That is, they must develop a new kind of epistemic agency that is not defined by independence from technology, but by the capacity to navigate and regulate epistemic entanglement with it. Existing learning theories provide limited support for this task.

3. Beyond individual cognition: rethinking epistemic activity in hybrid learning systems

The epistemic shifts introduced by GenAI necessitate a reconceptualization of agency, reasoning, and responsibility that existing learning theories cannot accommodate. Distributed cognition, sociomaterialism, and posthumanist theory reframe epistemic activity as emergent, relational, and enacted within sociotechnical systems.

Distributed cognition (Hollan et al., 2000) challenges the assumption that thinking resides within individuals. Instead, it emphasizes that cognition unfolds across people, artifacts, tools, and environments. In this view, reasoning is not solely an internal act but a coordinated process involving external representations such as diagrams, checklists. For example, a medical student's diagnostic reasoning is not confined to what they remember from coursework or lectures. Instead, it emerges through interaction with a distributed system of cognitive supports, such as the electronic health record (EHR) to review the patient's prior history and lab values, and a decision-support tool to estimate the probability of a pulmonary embolism (Boyle et al., 2023).

Similarly, when learners engage with GenAI, they are not merely retrieving information; they are participating in a distributed cognitive system in which the AI's outputs shape their attention, prompt inference, and scaffold explanation. Empirical analyses of student engagement further illustrate this distributed positioning. Fang and Zhou (2025) document interaction patterns ranging from passive acceptance of AI-generated content to dialogic co-learning postures, suggesting that epistemic participation varies substantially across learners within AI-mediated reasoning environments. While GenAI lacks intentionality, it functions as an active component, a “co-learner” in learners' reasoning environments engaging “in dialogue, [providing] feedback, and [co-constructing] understanding with the learner” (Fang & Zhou, 2025). While distributed cognition treats tools as cognitive components, all tools are treated equally, be it a checklist or a GenAI system.

While distributed cognition treats tools and artifacts as components of cognitive systems, it does not centrally theorize how learners evaluate or position themselves toward epistemic risk when system-generated information is incorrect or misleading. Human participants may and should engage in such interrogation, but the framework offers limited analytic vocabulary for examining how responsibility for evaluating generated knowledge is negotiated within human–AI interaction. Nor does it provide guidance on how to navigate this uncertainty. This is where sociomaterialist theory offers critical insights.

Sociomaterialism refuses the binary separation between social and material actors (Bavdas, 2018). It treats knowledge as emerging through entangled relations between humans and nonhumans where tools, discourses, technologies, and institutional structures co-produce meaning and practice. For example, an architecture student's design reasoning is not confined to internal visualization or abstract conceptualization. Instead, it emerges through interaction with a sociomaterial system of practice: sketching tools that constrain line precision, model-making materials that resist certain forms, peer critiques that introduce alternative framings, and spatial norms of the studio that shape how work is displayed and interpreted.

From this perspective, GenAI is not simply a support mechanism; it is a constitutive part of the learning environment, shaping not only what is known but also how knowledge is performed, negotiated, and stabilized.

When learners rely on AI to generate arguments or reword their responses, they are engaging in sociomaterial practices in which agency is not possessed but emerges through interaction. But even sociomaterialism does not fully explain what happens when those systems start generating ideas, framing arguments, or simulating conversation. For that, we need posthumanist theory.

Posthumanist theory decenters the human as the exclusive site of cognition or meaning-making. Posthumanism does not anthropomorphize machines, nor does it equate AI with human consciousness. Rather, it problematizes the assumption that agency and understanding are exclusive to humans (Forlano, 2017). For example, a learner using a virtual simulation to explore climate change models adjusts variables like CO₂ emissions or deforestation rates and watches as the system dynamically updates projected outcomes. The simulation responds in real time, offers causal explanations, and visualizes long-term planetary shifts, creating a learning experience that feels dialogic, even though no other human is present. The student comes to understand complex systems not by internal reasoning alone, but by interacting with a nonhuman entity that mimics explanatory behavior.

In AI-mediated environments, learners are not simply interacting with tools; they are operating within hybrid ecologies where machine outputs carry epistemic weight. Posthumanist perspectives help us recognize that the learner's role is no longer defined by independence from the system, but by their capacity to navigate, negotiate, and respond to it with epistemic reflexivity. The key question becomes not whether AI systems “know,” but how outputs that simulate knowing shape what learners come to accept, contest, or reconsider as knowledge.

Distributed cognition, sociomaterialism, and posthumanism shift our understanding of where and how knowledge is produced. But they also push us to rethink who the epistemic actor is in these environments. If knowledge is no longer located solely in the individual, and if tools now generate, reframe, and provoke ideas in real time, then epistemic agency itself must be reconceptualized. We need a model that captures not only the learner's capacity to think critically, but their ability to navigate, regulate, and respond within entangled human–AI systems. This requires moving beyond epistemic agency as an individual trait, toward what I call epistemic co-agency.

4. From epistemic agency to epistemic Co-agency

The challenge of GenAI is that learners must now exercise judgment in environments where knowledge is increasingly co-produced by systems they do not fully control or understand. This creates a learning environment where learners are no longer the sole epistemic actors in their own learning processes. Yet AI systems are not co-authors or collaborators in any symmetrical sense. They do not possess beliefs, intentions, or the capacity for self-reflection. Treating them as full partners in meaning-making would conflate simulation with cognition.

Epistemic co-agency, as I propose here, acknowledges this asymmetry. It draws on the concept of relational agency, where learning emerges through interaction with tools, symbols, and others within mediated activity systems. In this framing, agency is not an individual activity, but something enacted through coordination with external systems, including AI. The learner's epistemic posture evolves through ongoing interaction with AI, including how they respond to AI outputs, when they trust or resist them, and how they integrate them into broader reasoning. In this sense, GenAI becomes a constitutive part of the learning environment, shaping not only what is known but how it is known.

Epistemic co-agency distances itself from metaphors like “AI partners” or “co-authors,” which imply mutuality and shared intention. Instead, epistemic co-agency focuses on the learner's capacity to maintain epistemic sovereignty, thereby interrogating, contextualizing, and regulating the influence of AI within their own knowledge practices. Furthermore, epistemic co-agency treats the learner's engagement with

AI not as a fixed trait, but as a developmental process that unfolds through experience, reflection, and guided interaction. It calls for learning environments that do more than deliver content or enforce safeguards; they must help learners see how AI generates claims, understand its limits, and respond with epistemic reflexivity.

To support this developmental view of learner–AI engagement, I propose the Epistemic Entanglement Framework that captures how reasoning unfolds across different configurations of human–AI interaction. It describes three distinct configurations: Epistemic Augmentation, Epistemic Integration, and Epistemic Co-Agency. Each configuration represents a distinct way learners interact with AI-generated knowledge. These configurations are not stages of maturity, but patterned epistemic relationships that vary by task, design, and learner posture. What follows is a detailed articulation of the framework and its distinct human posture, cognitive demand, and design implication.

5. Epistemic Entanglement Framework

Human–AI interactions can be understood through three epistemic configurations: Epistemic Augmentation (EA), Epistemic Integration (EI), and Epistemic Co-Agency (EC). These configurations describe interactions between learners and AI-generated knowledge within entangled reasoning environments. They are not stages of development or levels of proficiency. Rather, they represent distinct epistemic postures differentiated by how learners assume responsibility for evaluating and justifying knowledge claims. See Fig. 1.

5.1. Epistemic Augmentation (EA)

Epistemic Augmentation (EA) treats the AI as a content generator. Learners assume a low-interrogation posture and use GenAI to retrieve summaries, draft paragraphs, or generate concept explanations without engaging in recursive thought or epistemic transformation. While this may improve surface-level productivity, it does not alter how knowledge is constructed or evaluated. The learner remains epistemically passive, outsourcing cognition without reflection.

Within this configuration, learners engage AI outputs as useable material rather than as knowledge claims requiring interrogation. Generated responses are accepted, adapted, or reproduced with limited examination of their assumptions, authority, or limitations. Epistemic positioning remains oriented toward application and incorporation, with evaluation and justification of claims only minimally foregrounded.

Epistemic Augmentation does not imply deficiency or misuse. It reflects a common and often productive orientation when efficiency, exposure to content, or initial comprehension are the primary goals. However, because generated knowledge is treated primarily as consumable input, the configuration offers limited opportunity for learners to actively position themselves toward the epistemic status of claims emerging through interaction.

5.2. Epistemic Integration (EI)

Epistemic Integration (EI) configures human–AI interaction as a recursive, co-structured process in which learners strategically prompt the AI, refine outputs, and integrate them into evolving reasoning. Multiple sources of epistemic input are coordinated, including personal judgment, peer perspectives, and generated contributions. Within this configuration, learners position themselves as managers of AI-assisted reasoning: outputs are interpreted, modified, compared, or selectively adopted to improve coherence, relevance, or disciplinary alignment.

AI is not treated as a passive answer engine but as a tool whose contributions shape both product and process. Through iterative negotiation—modifying prompts, probing assumptions, and comparing outputs with disciplinary norms or human feedback—the learner actively regulates the interaction. Boundaries between user and system become permeable as reasoning is co-shaped through exchange.

Despite this active engagement, epistemic positioning remains oriented toward managing knowledge production rather than interrogating knowledge claims themselves. Generated contributions are refined and integrated, but their authority, assumptions, or warrant typically remain tacit. The learner is epistemically active, while the AI's epistemic influence remains largely unexamined.

5.3. Epistemic Co-Agency (EC)

Epistemic Co-Agency characterizes configurations in which learners position themselves not only as managers of AI-assisted reasoning but also as accountable evaluators of the knowledge claims that emerge through interaction. AI outputs are treated as epistemically contestable contributions, subject to critique, reframing, or rejection, rather than as material to be incorporated or optimized.

AI outputs are engaged dialectically—as prompts to surface contradictions, challenge assumptions, and reframe perspectives. Meaning emerges through frictional, dialogic exchange, where the learner maintains epistemic sovereignty while navigating the cognitive

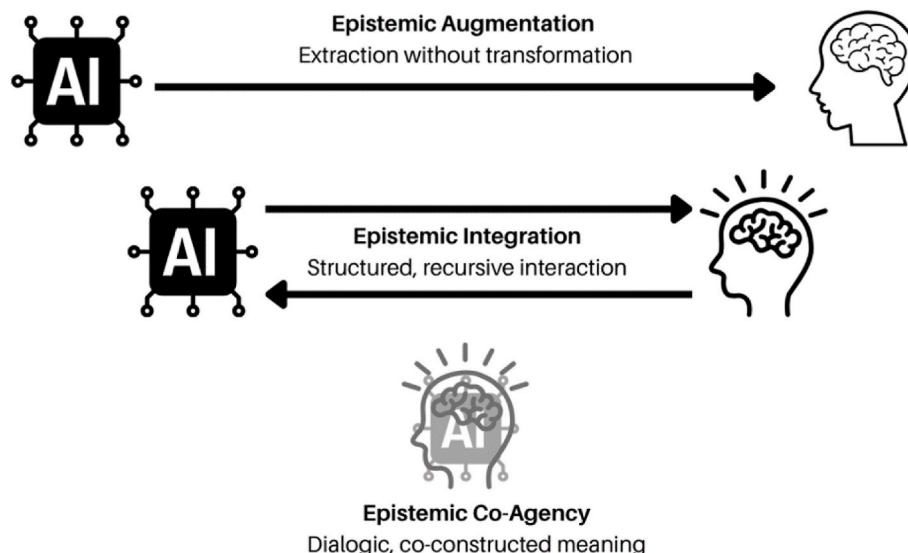


Fig. 1. Epistemic entanglement framework.

provocations introduced by the system. Engagement is oriented toward evaluating validity, contextualizing meaning, and justifying acceptance or refusal. AI participation shapes the reasoning environment, but epistemic authority remains located with the learner as knower.

This positioning aligns with accounts of distributed authorship in which generative contributions may restructure reasoning processes while evaluative responsibility remains situated with the human participant who selects, interprets, and warrants the resulting claims (Rodrigues, 2025). Co-agency thus reflects not shared cognition but relational accountability enacted within hybrid epistemic systems. Co-agency does not imply symmetry of agency between human and system. Generative AI neither intends nor understands; its participation lies in shaping possibilities for reasoning. Epistemic co-agency instead refers to a relational configuration in which learners maintain sovereignty over judgment while reasoning within an entangled system of influence and response.

This configuration reflects a shift in epistemic positioning from managing knowledge production to evaluating and warranting knowledge itself. The defining feature of Epistemic Co-Agency is therefore not interactional complexity but the learner's explicit orientation toward the status, limits, and justification of claims that emerge from human–AI engagement.

5.4. EA-EI-EC configurations

To situate the framework within existing scholarship on human–technology agency, the configurations may also be described using a set of analytic dimensions informed by Thürmel's (2015) graduated agency framework. These dimensions are not intended to define or differentiate the configurations, which are primarily distinguished by the location of epistemic responsibility, but rather to provide interpretive lenses for examining patterns of interaction across human–AI systems.

The five dimensions are: activity level, adaptivity, interaction potential, personification, and joint agency. Activity level refers to the extent to which AI outputs shape or redirect the interaction. Adaptivity describes how responsively outputs shift in relation to learner input. Interaction potential captures the depth and iterative complexity of exchange. Personification concerns the extent to which learners attribute perspectival qualities to the system. Joint agency reflects how outcomes appear co-produced within the interaction.

These dimensions are offered as descriptive scaffolding to connect the epistemic entanglement framework with existing agency theory traditions and to support future analytic work, rather than as scalar measures or defining components of the configurations themselves. See Fig. 2.

While Epistemic Co-Agency foregrounds explicit engagement with the status and justification of knowledge claims, Epistemic Augmentation and Integration are not inherently deficient (see Fig. 3). They may align with particular task conditions, time constraints, or learning goals. The concern lies in the unexamined assumption that any single

configuration is universally sufficient. The task of learning design is not to prescribe EC in all cases, but to help learners recognize which configuration they are in, understand its limits, and develop the agility to shift epistemic postures as needed.

6. Pedagogical affordances and epistemic limits across configurations

All three configurations, EA, EI, and EC serve distinct pedagogical purposes, depending on the task demands, learner readiness, and instructional goals. The value of the framework lies in making configurations visible as objects of reflection rather than prescribing movement among them. By foregrounding epistemic awareness, the framework supports pedagogical approaches that emphasize situational judgment, adaptive engagement, and accountability in human–AI knowledge construction.

6.1. EA: instrumental access and provisional scaffolding

Epistemic Augmentation (EA) often serves as a necessary entry point for learners unfamiliar with generative AI or those engaging in foundational tasks that prioritize efficiency, comprehension, or exposure to new content. It provides immediate value in helping students access explanations, clarify concepts, and scaffold basic understanding. In large introductory courses, for instance, EA may serve as a low-barrier support mechanism, ensuring students do not stall at the threshold of a problem. For time-sensitive or high-volume tasks—such as summarizing readings, formulating draft outlines, or reviewing terminology—EA can streamline the learning process without sacrificing core goals.

EA can provide functional value when learners are building preliminary schema, encountering unfamiliar terminology, or engaging in exploratory exposure to new domains where deep critique is not yet feasible. Early-stage learners cannot interrogate what they don't yet conceptually grasp (Kirschner et al., 2010). Under high task load or early-stage comprehension demands, outsourcing explanatory generation allows learners to preserve cognitive resources for later reasoning or integration. In such contexts, AI outputs serve as provisional scaffolding rather than epistemic endpoints.

6.2. EI: recursive coordination and strategic refinement

Epistemic Integration (EI) comes into play when learners begin to use GenAI not just as a source of answers, but as a mechanism for shaping their own responses. It supports recursive engagement, strategic prompting, and iterative refinement of ideas. In writing-intensive courses, design studios, or research settings, EI can scaffold the development of structure, tone, and disciplinary framing. Learners operating in EI are still epistemically active, but they often rely on AI to amplify or refine rather than challenge their own thinking.

Within this configuration, AI participation can extend cognitive reach by enabling rapid iteration, perspective comparison, and structural experimentation that might otherwise be constrained by time or working-memory limitations. Learners can test alternative framings, surface latent connections, or refine expression through cycles of interaction. Such activity may strengthen metacognitive awareness of process and improve strategic engagement with knowledge construction, even when underlying epistemic warrants remain implicit.

6.3. Developmental risks of configuration persistence

While Epistemic Augmentation (EA) and Epistemic Integration (EI) have legitimate roles in learning, remaining in either mode for too long carries risks of epistemic underdevelopment. When learners stay in EA, they grow accustomed to outsourcing cognitive labor. Over time, this fosters a posture of epistemic passivity (Fang & Zhou, 2025), where AI outputs are accepted as authoritative, even when they are flawed or

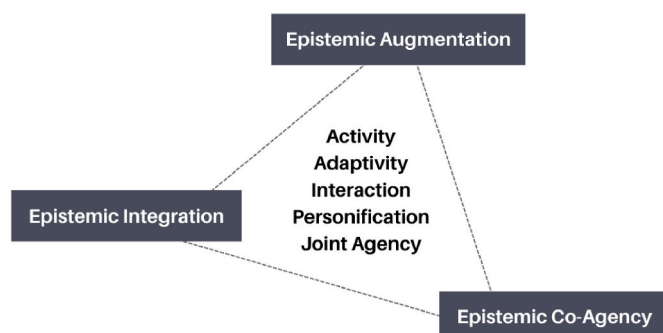


Fig. 2. Epistemic activity across the epistemic entanglement framework.

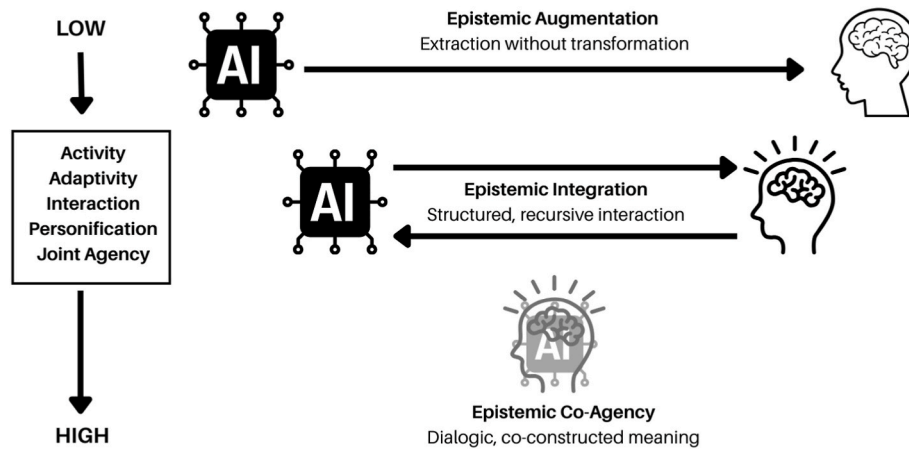


Fig. 3. Epistemic activity levels across the epistemic entanglement framework.

incomplete. Such interactions are reinforced by the persuasive fluency of generated text, where stylistic coherence can simulate epistemic adequacy without corresponding justificatory grounding (Rodrigues, 2025). Within EA configurations, this characteristic amplifies the likelihood that efficiency is mistaken for understanding.

Because EA interactions often feel efficient and productive, learners may mistakenly equate fluency of output with depth of understanding. The result is a subtle erosion of epistemic vigilance. Learners begin to defer to AI not because it earns trust, but because it is fast, fluent, and always available. They lose the habit of questioning, of examining evidence, and of acknowledging uncertainty. Over time, reasoning becomes more procedural than reflective. For this reason, EA is best understood as situationally appropriate but developmentally incomplete. The purpose of identifying this configuration is not to endorse passive uptake, but to render visible a prevalent interaction posture so that instructional design can explicitly address its affordances and risks.

EI, while more sophisticated, also presents hidden risks. EI is bounded by its typical orientation toward optimization rather than interrogation. While learners actively shape outputs, they often do not examine the epistemic assumptions embedded within generated contributions. Authority signals, omissions, or biases may remain tacit. Consequently, prolonged reliance on EI may cultivate procedural fluency without corresponding development of evaluative judgment. Identifying EI as a distinct configuration therefore allows instructional design to leverage its productive affordances while recognizing the need to scaffold movement toward deeper epistemic reflexivity when task demands require it.

From a learning design perspective, the goal is not to prescribe Epistemic Co-Agency as a universal endpoint, but to cultivate learners' awareness of their current epistemic positioning and its implications. Instruction can support learners in recognizing how they are engaging AI-generated contributions, understanding the affordances and limitations associated with that configuration, and developing the flexibility to reposition themselves when epistemic demands shift.

The three configurations - EA, EI, EC - are not fixed tracks or ability levels; they are fluid stances that shift based on context, task, and design. Effective instructional environments must support movement across the three configurations and teach students how to recognize when to shift stances—when to use AI for efficiency, when to slow down for reflection, and when to lean into uncertainty. Although Epistemic Co-Agency foregrounds evaluative engagement with knowledge claims, the framework does not assume that such positioning is universally required. Epistemic effort is situationally allocated across tasks, contexts, and cognitive demands. Many learning activities involve instrumental or coordinative goals where sustained interrogation of generated content may neither be necessary nor productive. The configurations, therefore, represent dominant epistemic orientations rather than

hierarchical stages, and elements of evaluative judgment may be present across them.

7. Situated illustrations of epistemic configuration enactment

This section presents illustrative enactments from health professions education and engineering to clarify how the Epistemic Entanglement Framework may operate in authentic educational contexts. These scenarios are not empirical case studies but analytically grounded examples that demonstrate how epistemic configurations manifest in situated learning environments and how learners may shift their postures in response to task demands.

7.1. Health professions education

Consider a medical student preparing for clinical rounds using a generative AI system to review pathophysiology. In an Epistemic Augmentation configuration, the learner may retrieve summaries of diagnostic criteria or pharmacological mechanisms to establish foundational orientation. The generated explanations function as provisional scaffolds, supporting schema activation without requiring immediate interrogation. In this context, efficiency and cognitive load management are aligned with instructional goals centered on preparation rather than evaluation.

As the learner engages in case-based reasoning, interaction may shift toward Epistemic Integration. The student iteratively prompts the system to explore differential diagnoses, compares generated rationales with clinical guidelines, and refines their understanding through recursive exchange. Here, AI participation extends the learner's representational workspace, enabling rapid iteration across diagnostic possibilities. Epistemic activity centers on coordinating multiple knowledge inputs rather than interrogating their warrant.

In simulation or supervised clinical debriefing, instructional design may deliberately foreground Epistemic Co-Agency. Learners critique AI-suggested treatment plans, identify omissions relative to patient context, and justify acceptance or rejection of recommendations using disciplinary evidence. Engagement becomes dialogic and evaluative, with epistemic responsibility explicitly retained by the learner. The instructional objective shifts from information management to epistemic judgment under uncertainty.

These transitions illustrate how configurations are not stages of ability but situated postures shaped by task structure, stakes, and pedagogical intent.

7.2. Engineering design and computational modeling

A contrasting illustration can be drawn from engineering education,

where learners engage with generative AI systems during design or modeling tasks. In early-stage project work, a student tasked with developing a structural or computational solution may operate within Epistemic Augmentation by retrieving summaries of material properties, algorithmic structures, or regulatory standards. AI outputs serve as rapid orientation resources that support entry into the design space without requiring immediate epistemic evaluation. In this configuration, interaction prioritizes efficiency and domain familiarization.

As design development progresses, interaction often shifts toward Epistemic Integration. Learners iteratively prompt AI systems to generate alternative design configurations, test modeling approaches, or explore parameter variations. Outputs are compared, adapted, and incorporated into evolving prototypes or simulations. AI participation expands the solution landscape, allowing learners to examine structural trade-offs or performance implications through recursive interaction. Epistemic activity centers on coordinating generated possibilities with disciplinary constraints rather than interrogating the epistemic assumptions embedded in the outputs themselves.

Instructional interventions such as design reviews or failure-analysis exercises may intentionally foreground Epistemic Co-Agency. Students critically evaluate AI-generated design rationales, identify unrealistic assumptions, and justify modifications using engineering principles or empirical validation. Here, engagement becomes evaluative and accountable, as learners position themselves as responsible arbiters of system reliability and safety. The emphasis shifts from generative exploration to epistemic justification grounded in disciplinary evidence.

This scenario highlights how epistemic configurations emerge differently within technical problem-solving contexts compared to interpretive or professional domains. The framework accommodates these disciplinary distinctions by treating configurations as situated epistemic orientations shaped by task structure, validation norms, and instructional framing rather than as uniform learning stages.

7.3. Implications for instructional integration

Across these contexts, the framework demonstrates that epistemic configurations are enacted through instructional design rather than solely learner disposition. Tasks emphasizing preparation, iteration, or evaluation selectively foreground different epistemic orientations. Recognizing these configurations allows educators to structure AI-mediated activities that intentionally shift epistemic demands rather than assuming uniform critical engagement across all interactions.

Situating the framework within concrete educational settings thus illustrates its utility as an analytic and design lens. It enables instructors to identify dominant epistemic postures within learning activities, anticipate associated affordances and risks, and design transitions that cultivate adaptive epistemic positioning in AI-mediated environments.

8. Designing for Epistemic Co-Agency: instructional and assessment implications

Designing learning environments for AI-mediated reasoning demands a reorientation of pedagogy itself. The goal is not simply to teach students how to use AI more effectively, but to support their development as epistemic co-agents who can think with, through, and against generative systems. This shift has deep implications for both instructional design and assessment practices.

8.1. Instructional design: scaffolding toward Co-Agency

Each epistemic configuration, EA, EI, and EC, requires different forms of support. In the early stages, design should focus on epistemic awareness: helping students recognize what AI is doing, how it generates responses, and when its output may be misleading or limited. Prompts, reflection checklists, and structured comparison tasks can guide learners through these foundational stages (EA and EI) without allowing them to

stall there.

However, the ultimate goal is not the mastery of AI as a tool, but the cultivation of learners who can treat AI outputs as dialogic material. To reach EC, the design must prioritize epistemic reflexivity. This means constructing learning experiences that embed contradiction, require students to take epistemic stances, and push them to reframe AI responses as starting points rather than endpoints. This is not just design for skill acquisition, but design for epistemic maturation.

Many current AI-in-education tools prioritize efficiency, aiming to deliver faster feedback (SchoolAI, Curipod), more precise explanations (ChatGPT), or streamlined workflows (ChatGPT). While these goals are legitimate, they risk reinforcing passive epistemic postures that encourage learners to outsource reasoning and overtrust system outputs. To support Epistemic Co-Agency, systems must embed epistemic friction or features that prompt learners to reflect, question, or reconsider AI outputs.

This can include:

- Prompt templates that ask learners to explain why an AI output is appropriate or flawed
- Multiple-output displays that surface divergent AI responses to the same prompt
- Confidence visualizations or traceable sources to signal uncertainty or justification
- Embedded critique scaffolds, such as “What’s missing?”, “What’s contested?”, or “Whose perspective is absent?”

These features do not slow learning—they deepen it. They align with design-for-learning principles that treat interface and functionality as epistemic tools, not just delivery mechanisms.

Comparable instructional approaches have begun to emerge in empirical classroom settings, where structured reflection and comparative evaluation tasks have been used to prompt students to examine their positioning toward AI-generated contributions rather than treating outputs as transparent solutions (Fang & Zhou, 2025). Such designs illustrate how epistemic awareness can be operationalized without presuming uniform critical engagement.

8.2. Assessment: measuring epistemic trajectory, not just output

Traditional assessments that focus on correctness, coherence, or completeness are poorly suited to capture the kinds of epistemic shifts the epistemic entanglement framework targets. What’s needed are assessments that attend to how learners engage with knowledge, not just what they produce.

At the EA level, assessment might focus on awareness: Can the student identify the source of an AI-generated claim? Do they recognize when the output needs modification or contextualization? In EI, assessment should examine strategic integration: How well does the student coordinate AI input with other sources? Can they triangulate perspectives or detect epistemic gaps? At the EC level, assessment must capture epistemic reflexivity and transformation: Does the student interrogate assumptions? Reframe their own thinking? Use the AI to deepen—not shortcut—their reasoning?

To support this, assessments should include reflective prompts, explanations of the process, and evidence of revision. Rubrics might evaluate not only the final artifact, but the reasoning trajectory: how the learner moved from question to insight, and how AI figured into that arc. Importantly, such assessment designs also need to protect against superficial “AI detection” measures that reduce engagement to surveillance, rather than fostering deeper epistemic agency.

9. Limits of AI literacy and safeguards

Much of the current educational response to GenAI has focused on what is often referred to as “AI literacy,” or teaching students how these

systems work, their limitations, and how to use them responsibly (Chiu et al., 2024). Others have focused on technical safeguards, such as embedding plagiarism detection tools, restricting access to AI during assessments, or providing policy guidelines for the ethical use of AI (Ogwueleka, 2025; Pudasaini et al., 2024). These are important first steps. But on their own, they don't address the deeper educational question of how learners develop the ability to reason with, question, and take responsibility for knowledge in environments where AI now helps shape what counts as "an answer."

GenAI is not just a technical addition. It is a new epistemic actor in the learning environment. Institutions must recognize that decisions about AI use are also decisions about what kinds of knowledge practices are being cultivated. This means GenAI tools cannot be framed simply as "assistants," but as objects of inquiry: What are their limitations? What values are encoded in their design? Discussions of epistemic ethics, source reliability, and knowledge construction must be embedded into disciplinary curricula—not just standalone "AI literacy" modules. And faculty must be supported in developing the confidence and capacity to teach in AI-mediated environments, especially when disciplinary norms vary in how knowledge is validated (Conceição & Samuel, 2025).

Ultimately, designing for Epistemic Co-Agency requires aligning technological design, instructional practice, and institutional norms toward a common goal: cultivating learners who can navigate complex, hybrid knowledge ecologies with critical insight and epistemic responsibility. This paper provides a conceptual foundation for designing AI-mediated learning environments that support epistemic development; however, empirical research is necessary to test and refine this proposed model. Future studies may investigate how learners progress through co-agency stages in various disciplinary contexts, how specific design features impact epistemic behavior over time, and how institutional or cultural norms influence learner–AI interaction. There is also a need to align these frameworks with emerging assessment practices, faculty development efforts, and policy debates about the role of AI in education. As generative technologies continue to evolve, research must remain attentive not only to their capabilities but to their impact on the formation of learners as epistemic agents.

10. Conclusion

The integration of GenAI into higher education raises urgent questions not only about learning outcomes or academic integrity, but about the future of epistemic development. As learners increasingly encounter systems that generate fluent, persuasive, and seemingly authoritative responses, the risk is not merely misuse but epistemic displacement: a gradual erosion of learners' capacity to question, evaluate, and take responsibility for knowledge.

In this paper, I have argued that addressing this challenge requires more than AI literacy or technical safeguards. It demands a shift in how we design learning environments, structure learner–AI interaction, and conceptualize the learner's evolving epistemic posture. The Epistemic Entanglement Framework reframes the learner–AI relationship not as a static tool-use paradigm or naive collaboration, but as a developmental process in which learners come to navigate, resist, and repurpose AI-generated knowledge with increasing reflexivity and agency.

The educational stakes are high. Without intentional design, GenAI risks flattening learners' epistemic development, encouraging passive acceptance over active inquiry. But with deliberate scaffolding, ethical framing, and instructional integration, these systems can instead become provocative companions—catalysts for deeper thinking, more responsible judgment, and renewed ownership of knowledge.

What is ultimately at stake is not just how students use AI, but who they become in relation to it. In reclaiming epistemic development as a central concern of AI-in-education design, we affirm that education is not merely about accessing information or acquiring skills, but about cultivating critical, capable, and agentic knowers in an increasingly intelligent world.

Disclaimer

The opinions and assertions expressed herein are those of the author (s) and do not necessarily reflect the official policy or position of the Uniformed Services University or the Department of War.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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The system did not generate theoretical constructs, analytic claims, or interpretations adopted without author evaluation. All substantive conceptual decisions, argument development, and final wording were determined and reviewed by the authors, who assume full responsibility for the content. No AI-generated text was incorporated without critical revision, and all references and citations were independently verified for accuracy and relevance.

ChatGPT-4o and 5.2 was accessed with training mode turned off and temporary chat enabled, ensuring that interactions were not stored or used to improve the model.

ChatGPT-4o and 5.2 served as a cognitive partner to support outlining, language enhancement, and critical reflection. The author retained full control over the scholarly content, and all output was independently reviewed and revised to uphold academic integrity and authorship standards.

References

- Alvarado, R. (2023). AI as an epistemic technology. *Science and Engineering Ethics*, 29(5). <https://doi.org/10.1007/s11948-023-00451-3>
- Bavdaz, A. (2018). Past and recent conceptualisations of sociomateriality and its features. *Athens Journal of Social Sciences*, 5(1), 51–78.
- Bernabei, M., Colabianchi, S., Falegnami, A., & Costantino, F. (2023). Students' use of large language models in engineering education: A case study on technology acceptance, perceptions, efficacy, and detection chances. *Computers and Education: Artificial Intelligence*, 5, Article 100172.
- Boyle, J. G., Walters, M. R., Jamieson, S., & Durning, S. J. (2023). Distributed cognition: Theoretical insights and practical applications to health professions education: AMEE guide no. 159. *Medical Teacher*, 45(12), 1323–1333.
- Chen, J., Fife, J. H., Bejar, I. I., & Rupp, A. A. (2016). Building e-rater® scoring models using machine learning methods. *ETS Research Report Series*, 2016(1), 1–12.
- Chiu, T. K., Ahmad, Z., Ismailov, M., & Sanusi, I. T. (2024). What are artificial intelligence literacy and competency? A comprehensive framework to support them. *Computers and Education Open*, 6, Article 100171.
- Clark, N., Shen, H., Howe, B., & Mitra, T. (2025). Epistemic alignment: A mediating framework for User-LLM knowledge delivery. *arXiv preprint arXiv:2504.01205*.
- Conceição, S. C., & Samuel, A. (2025). Strategic integration of artificial intelligence in faculty development grounded in adult learning principles. *New Directions for Adult and Continuing Education*, 2025(187), 64–71.
- Fang, L., & Zhou, X. (2025). From tool to Co-Learner: Exploring student engagement with GenAI through the lens of social constructivism. *Evaluation Review*, Article 0193841X251411618.
- Forlano, L. (2017). Posthumanism and design. *She Ji: The journal of design, economics, and Innovation*, 3(1), 16–29.
- Grande, V., Kiesler, N., & Francisco, R. M. A. (2024). Student perspectives on using a large language model (llm) for an assignment on professional ethics. *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V.*, 1, 478–484.
- Hauswald, R. (2025). Artificial epistemic authorities. *Social Epistemology*. <https://doi.org/10.1080/02691728.2025.2449602>
- Hollan, J., Hutchins, E., & Kirsh, D. (2000). Distributed cognition: Toward a new foundation for human-computer interaction research. *ACM Transactions on Computer-Human Interaction*, 7(2), 174–196.
- Hutchins, E. (2000). Distributed cognition. *International encyclopedia of the social and behavioral sciences*, 138(1), 1–10.
- Kirschner, P. A., Sweller, J., Clark, R. E., Kirschner, P. A., & Clark, R. E. (2010). *Why minimal guidance during instruction does not work: An analysis of the failure of*

- constructivist. *Based teaching Work: An Analysis of the Failure of Constructivist, Discovery, Problem-Based, Experiential, and Inquiry-Based Teaching* (November 2014), 37–41.
- Kozulin, A. (2002). Sociocultural theory and the mediated learning experience. *School Psychology International*, 23(1), 7–35.
- Miller, E., Manz, E., Russ, R., Stroupe, D., & Berland, L. (2018). Addressing the epistemic elephant in the room: Epistemic agency and the next generation science standards. *Journal of Research in Science Teaching*, 55(7), 1053–1075. <https://doi.org/10.1002/tea.21459>
- Nieminen, J. H., & Ketonen, L. (2024). Epistemic agency: A link between assessment, knowledge and society. *Higher Education*, 88(2), 777–794.
- Ogwueleka, F. N. (2025). Plagiarism Detection in the age of artificial intelligence: Current technologies and future directions. In *AI and ethics, academic integrity and the future of quality assurance in higher education* (Vol. 10).
- Oh, P. S., & Lee, G. G. (2024). Confronting imminent challenges in humane epistemic agency in science education: An interview with ChatGPT. *Science and Education*. <https://doi.org/10.1007/s11191-024-00515-1>
- O'Neill, R., & Russell, A. (2019). Stop! grammar time: University students' perceptions of the automated feedback program Grammarly. *Australasian Journal of Educational Technology*, 35(1).
- Pudasaini, S., Miralles-Pechuán, L., Lillis, D., & Llorens Salvador, M. (2024). Survey on AI-generated plagiarism detection: The impact of large language models on academic integrity. *Journal of Academic Ethics*, 1–34.
- Rodrigues, T. V. (2025). Distant writing and the epistemology of authorship: On creativity, delegation, and plagiarism in the age of AI. *International Journal of Social Sciences and Humanities Invention*, 12(5), 8598–8613. <https://philpapers.org/archive/RODDWA.pdf>.
- Rosenberg, L. B. (2023). *The manipulation problem: Conversational AI as a threat to epistemic agency*. Association for Computing Machinery.
- Siemens, G. (2017). *Connectivism. Foundations of learning and instructional design technology*.
- Sjøberg, S. (2010). Constructivism and learning. *International Encyclopedia of Education*, 5, 485–490.
- Stroupe, D. (2014). Examining classroom science practice communities: How teachers and students negotiate epistemic agency and learn science-as-practice. *Science Education*, 98(3), 487–516. <https://doi.org/10.1002/scs.21112>
- Thürmel, S. (2015). The participatory turn: A multidimensional gradual agency concept for human and non-human actors. In *Collective agency and cooperation in natural and artificial systems: Explanation, implementation and simulation* (pp. 45–60). Cham: Springer International Publishing.
- Wu, J. Y., Lee, Y. H., Chai, C. S., & Tsai, C. C. (2025). Strengthening human epistemic agency in the symbiotic learning partnership with generative artificial intelligence. *Educational Researcher*, Article 0013189X251333628.
- Zhang, Z., Gao, J., Dhaliwal, R. S., & Li, T. J. J. (2023). Visar: A human-ai argumentative writing assistant with visual programming and rapid draft prototyping. In *Proceedings of the 36th annual ACM symposium on user interface software and technology* (pp. 1–30).